BSIT MELEC 9: System Need Analysis

Module 3 Study Reviewer & Practice Examination

COURSE: BSIT - MELEC 9 (System Need Analysis)	**REFERENCE:** BSIT_LS_MELEC9_N03.docx (Learning Sheet No. 3)
TOPIC: Project Planning and Feasibility Study	**COVERAGE:** Prelim Examination
Key Competencies: Master the 6 components of a project plan, project team roles, and the complete TELOS feasibility evaluation framework.	**Format:** Notes + Practice Test + Answer Key

I. Understanding Project Planning

Project Planning is the foundational process of organizing, estimating, and scheduling everything needed before software construction begins. It serves as the master **roadmap** that guides developers and management throughout the project lifecycle.

Fundamental Planning Inquiries:
• *What* specific system will be built? | *Who* will develop, test, and manage it?
• *How much* capital is required? | *How long* will development take? | *What resources* are needed?

II. Six (6) Core Components of a Project Plan

Component	Core Purpose & Definition	School Enrollment System Example
1. Project Objectives	The explicit, high-level business goals and deliverables of the initiative.	Develop and deploy a cloud-based Online Enrollment and Grading System for the university.
2. Scope	Defines the precise boundaries: what is explicitly INCLUDED and what is EXCLUDED.	**Included:** Student registration, subject enrollment, grade viewing. **Excluded:** Payroll, HR, and bookstore inventory.
3. Budget	Detailed financial estimates required to procure equipment, pay talent, and test.	Total: Php 800,000 (Php 500k dev salaries, Php 200k servers/hardware, Php 50k software, Php 50k testing).
4. Schedule (Timeline)	Sequential breakdown of phases with realistic durations and milestone deadlines.	Planning: 2 wks | Analysis: 3 wks | Design: 3 wks | Coding: 8 wks | Testing: 4 wks | Deploy: 2 wks (Total: 22 wks).
5. Team Members	Clear delegation of roles, tasks, and accountability across project personnel.	1 Project Manager, 1 System Analyst, 1 UI/UX Designer, 3 Programmers, 1 QA Tester.
6. Resources	Physical hardware, software platforms, and personnel essential for execution.	**Software:** VS Code, MySQL, Git, XAMPP | **Hardware:** Laptops, cloud server, test devices.

EXAM TIP / CRITICAL CONCEPT: THE SCOPE BOUNDARY & SCOPE CREEP

Clearly stating **Excluded Features** in the Project Plan is just as vital as stating Inclusions! Without defined boundaries, clients constantly request extra features mid-project—a disastrous problem called *Scope Creep* that causes massive delays and budget failures.

III. Project Team Roles and Responsibilities

Role Title	Primary Responsibilities on the Software Project
Project Manager (PM)	Oversees overall execution, tracks budget, enforces schedule deadlines, coordinates team, and manages client expectations.
System Analyst	Elicits user requirements, analyzes business processes, models data workflows (DFD/ERD), and clarifies specifications for developers.
System Designer	Creates architectural blueprints, designs user interfaces (UI/UX wireframes), database table structures, and data flow patterns.
Programmer / Developer	Writes executable source code, constructs databases, connects application APIs, and conducts routine unit tests.
Quality Tester	Designs test cases, performs integration/system tests, uncovers bugs and defects, and ensures software satisfies acceptance criteria.

IV. Feasibility Study: The TELOS Framework

A **Feasibility Study** is an analytical evaluation conducted to answer one decisive question: ***'Should we build this project?'*** It determines whether a proposed system is practical, cost-effective, technically attainable, and aligned with organizational objectives.

The standard industry framework for evaluating feasibility is **TELOS**, evaluating five (5) distinct dimensions:

TELOS Dimension	Guiding Evaluative Question	Evaluation Criteria & Real-World Example
T - Technical Feasibility	*'Can we build the system with available technology and expertise?'*	Assesses existing hardware, cloud servers, development languages, and whether staff possess the technical programming skills.
E - Economic Feasibility	*'Is the project financially sound and worth the investment?'*	Performs Cost-Benefit Analysis (CBA) and Return on Investment (ROI). (e.g. Spending Php 500,000 saves Php 900,000 over 5 years = Feasible).
L - Legal Feasibility	*'Does the project comply with existing laws and regulations?'*	Ensures strict compliance with the Data Privacy Act of 2012, intellectual property laws, software licenses, and employee privacy.
O - Operational Feasibility	*'Will end-users accept, adopt, and successfully use the system?'*	Assesses user resistance, ease of use, organizational change management, and whether staff require extensive training.
S - Schedule Feasibility	*'Can the project realistically be finished within the deadline?'*	Evaluates milestones, strict institutional deadlines (e.g. system must go live before the August semester begins), and manpower availability.

MODULE 3: SAMPLE EXAMINATION

Test your mastery of Module 3. Complete all parts before checking the Answer Key.

PART I: MULTIPLE CHOICE (10 Items)

Select the best answer for each question.

1. What is the fundamental question answered by a Feasibility Study prior to starting a software project?

A. 'How many lines of code can we write in one day?'
B. 'Should we build this project?'
C. 'Which programming font is most aesthetic?'
D. 'How many monitors does each developer need?'

2. An analyst checks whether the school's existing computer hardware and campus Wi-Fi can run a proposed cloud portal. Which feasibility type is this?

A. Legal Feasibility
B. Technical Feasibility
C. Economic Feasibility
D. Schedule Feasibility

3. Calculating Return on Investment (ROI) by comparing an initial cost of Php 500,000 against projected savings of Php 900,000 is evaluated under:

A. Operational Feasibility
B. Economic Feasibility
C. Technical Feasibility
D. Schedule Feasibility

4. Ensuring that a student portal strictly adheres to the Philippine Data Privacy Act of 2012 is a primary concern of:

A. Schedule Feasibility
B. Legal Feasibility
C. Technical Feasibility
D. Operational Feasibility

5. If teachers refuse to use a newly deployed grading system because they find it too difficult and complicated, which feasibility dimension failed?

A. Operational Feasibility
B. Legal Feasibility
C. Technical Feasibility
D. Economic Feasibility

6. A university mandates that the new enrollment portal must be 100% operational before the August semester opens. Assessing if this can be achieved is:

A. Technical Feasibility
B. Operational Feasibility
C. Schedule Feasibility
D. Economic Feasibility

7. In a Project Plan, clearly identifying that the system will handle 'Student Enrollment' but will NOT include 'Payroll' defines the project's:

A. Scope
B. Schedule
C. Budget
D. Resources

8. Which software team role is primarily responsible for tracking project milestones, controlling budgets, and coordinating team activities?

A. System Analyst
B. Database Administrator

C. Project Manager (PM)
D. QA Tester

9. Which tool is categorized as a Software Resource in a software development project plan?

A. Dell PowerEdge Server
B. MySQL Database / Visual Studio Code
C. Quality Assurance Tester
D. Core i7 Laptop

10. What undesirable phenomenon occurs when extra features are continuously added to a project without expanding budget or timeline?

A. Scope Creep
B. Refactoring
C. Regression Testing
D. Feasibility Expansion

PART II: TELOS CLASSIFICATION (5 Items)

Classify each scenario into its corresponding TELOS dimension (Technical, Economic, Legal, Operational, or Schedule).

11. Assessing whether the school has sufficient funds to hire three senior Python developers and buy enterprise cloud licenses: ____________________
12. Evaluating whether existing clinic staff have the basic computer literacy needed to operate an electronic health records system: ____________________
13. Checking if third-party open-source libraries used in the software require royalty fees or violate proprietary copyright agreements: ____________________
14. Investigating if the development team has enough specialized knowledge in biometric facial recognition algorithms: ____________________
15. Determining whether a 16-week timeline is adequate to complete coding, testing, and deployment before annual accreditation: ____________________

PART III: TRUE OR FALSE (5 Items)

Write TRUE if the statement is correct; write FALSE if the statement is incorrect.

16. A project can be technically feasible and economically attractive, yet still fail completely due to poor operational feasibility.
17. The project scope should only describe features that are included; documenting excluded features is considered bad practice.
18. Economic feasibility focuses solely on software purchase price and ignores developer salaries, hardware, and ongoing maintenance.
19. Legal feasibility includes ensuring that user data handling conforms to data privacy, confidentiality, and statutory guidelines.
20. Conducting a feasibility study guarantees that a project will have zero technical challenges during programming.

PART IV: CASE STUDY & SCENARIO ANALYSIS (2 Items)

Analyze the practical project scenario below and provide reasoned evaluations.

Question 21 (Feasibility Evaluation):
City General Hospital wants an AI-driven Patient Triage System. The estimated development cost is Php 1,200,000. Expected operational savings and reduced misdiagnosis claims are projected at Php 2,800,000 over four years. However, the senior nursing staff refuses to use tablets, and the hospital's Wi-Fi network is unstable in emergency wards.
Evaluate this project across (a) Economic Feasibility, (b) Technical Feasibility, and (c) Operational Feasibility. Would you recommend proceeding as-is? Explain why.

Question 22 (Project Planning):
Why is defining the **Project Scope** regarded as one of the most critical steps in project planning? Describe what happens to an IT project when the scope is ambiguous or poorly communicated.

MODULE 3: ANSWER KEY & DETAILED RATIONALES

Question 1. B ('Should we build this project?')
Rationale: A feasibility study establishes viability and risk before an organization commits money, personnel, and time.

Question 2. B (Technical Feasibility)
Rationale: Assessing infrastructure (Wi-Fi, servers, client hardware, programming tools) evaluates technical capabilities.

Question 3. B (Economic Feasibility)
Rationale: Cost-Benefit Analysis and ROI compare capital expenditure against financial and operational return.

Question 4. B (Legal Feasibility)
Rationale: Statutory privacy regulations, data protection acts, and licensing compliance fall under legal feasibility.

Question 5. A (Operational Feasibility)
Rationale: Operational feasibility determines whether organizational culture, staff acceptance, and workflows will support the system.

Question 6. C (Schedule Feasibility)
Rationale: Assessing whether completion deadlines can be met with available resources is schedule feasibility.

Question 7. A (Scope)
Rationale: The scope establishes explicit boundaries by declaring what is in-scope and what is out-of-scope.

Question 8. C (Project Manager - PM)
Rationale: The PM coordinates schedules, manages resource allocations, tracks milestones, and communicates with sponsors.

Question 9. B (MySQL Database / Visual Studio Code)
Rationale: IDEs, DBMSs, compilers, and version control systems are categorized as Software Resources.

Question 10. A (Scope Creep)
Rationale: Uncontrolled feature expansion without corresponding budget/schedule adjustments is termed Scope Creep.

PART II: TELOS CLASSIFICATION ANSWERS

11. Economic Feasibility — Evaluating financial budgets, hiring costs, and software licensing expenses.

12. Operational Feasibility — Evaluating staff skills, user acceptance, and organizational readiness.

13. Legal Feasibility — Evaluating software licensing terms, open-source compliance, and copyright laws.

14. Technical Feasibility — Assessing team expertise and algorithm technical capabilities.

15. Schedule Feasibility — Evaluating milestone pacing against strict calendar deadlines.

PART III: TRUE OR FALSE ANSWERS

16. TRUE — Even brilliant, profitable software will fail completely if users refuse to adopt it in their daily workflow.

17. FALSE — Explicitly listing excluded items is essential to prevent misunderstandings and scope creep.

18. FALSE — Economic feasibility must account for total cost of ownership (TCO) including salaries, maintenance, and training.

19. TRUE — Privacy laws (like the DPA of 2012) require mandatory legal feasibility scrutiny.

20. FALSE — A feasibility study evaluates risk and viability; it cannot eliminate normal programming challenges.

PART IV: CASE STUDY & SCENARIO SAMPLE ANSWERS

Model Answer for Question 21 (Hospital Triage System Evaluation):
• **Economic Feasibility: FEASIBLE.** Cost (Php 1.2M) vs Savings (Php 2.8M) yields a net benefit of Php 1.6M, representing an excellent Return on Investment (ROI).
• **Technical Feasibility: NOT CURRENTLY FEASIBLE.** Unstable emergency room Wi-Fi will cause system dropouts during life-or-death patient triage.
• **Operational Feasibility: NOT CURRENTLY FEASIBLE.** Active resistance from senior nursing staff guarantees low system adoption and workarounds.
Recommendation: DO NOT proceed as-is. The hospital must first upgrade emergency network infrastructure (Technical) and conduct change management, ergonomic tablet selection, and nurse training (Operational) before proceeding.

Model Answer for Question 22 (Importance of Project Scope):
Project Scope is critical because it establishes an explicit agreement between stakeholders and developers regarding project boundaries.
Consequences of Ambiguous Scope:
1. *Scope Creep:* Stakeholders continuously add new feature requests without extending deadlines or budget.
2. *Schedule Delays & Cost Overruns:* Developers waste hours building unnecessary features while core requirements remain unfinished.
3. *Disputed Deliverables:* At project delivery, clients may refuse payment because their unstated expectations were not met.